SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 - ASSESSMENT TOOL 2

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.

(BL-1, BL-2, BL-3)

Assessment Tool Weightage:

QUESTIONS: 10 **Total Marks: 100**

Annexure A

Concept Mapping

Code of Conduct:

I P.Sai Ganesh Reddy(192525111) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

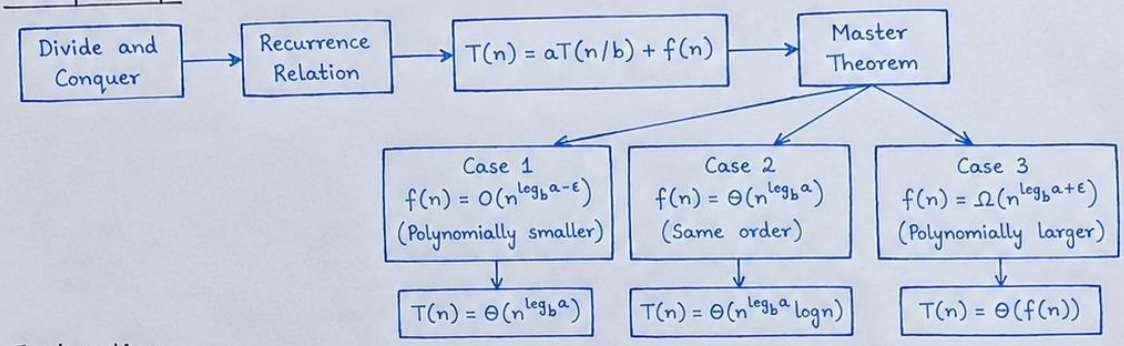
P.Sai Ganesh Reddy

RUBRICS FOR CALCULATION

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

3. Consider the concept map : Divide and Conquer → Recurrence Relation → $T(n) = aT(n/b) + f(n)$ → Master Theorem. Explain how each node leads to the next in the analysis of recursive algorithms. Extend the map by adding the three cases of the Master Theorem as child nodes and show how each case depends on the comparison between $f(n)$ and $n^{\log_b a}$. Justify how this mapping simplifies the process of deriving time complexity. Interpret how the completed map helps students analyze divide-and-conquer algorithms more efficiently.

Concept Map :



Explanation :

1. Divide and Conquer → The problem is divided into smaller subproblems, solved recursively and combined to get the final result.
2. Recurrence Relation → The time complexity of the recursive algorithm is expressed as a recurrence relation.
3. $T(n) = aT(n/b) + f(n)$ → This shows how the problem size reduces by a factor of b and the number of subproblems is a, with an additional non-recursive cost $f(n)$.
4. Master Theorem → Helps solve this type of recurrence by comparing $f(n)$ with $n^{\log_b a}$.
5. Three Cases → Based on the comparison, we get Case 1, Case 2 and Case 3, each giving a different form of time complexity.

Justification :

- The mapping simplifies the analysis because it provides a clear step-by-step flow from the problem structure to the final complexity.
- By comparing $f(n)$ with $n^{\log_b a}$, the Master Theorem directly gives the result without solving the recurrence manually.
- Each case covers different growth rates, so it helps in choosing the correct formula for the given recurrence.

Interpretation :

This concept map helps students analyze divide-and-conquer algorithms more efficiently because it breaks down the process into simple steps. It shows how the recurrence relation is formed and how the Master Theorem is applied. By understanding the three cases and their conditions, students can quickly identify which case fits the problem and derive the time complexity without complex calculations.